(Please write your Enrollment Number) Enrollment No. 12-701182021

End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(OffLine)
B.Tech (IT,CSE,MAE,ECE,DMAM, CSE-AI, ECE-AI) (Regular & Reappear) , Semester:3
(Nov-Dec, 2022)

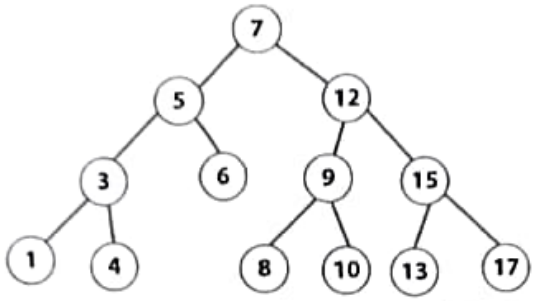
Subject Code: BCS 201 Subject: Data Structures
Time :3 Hours Maximum Marks :60
Note: Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1 (5*4=20)

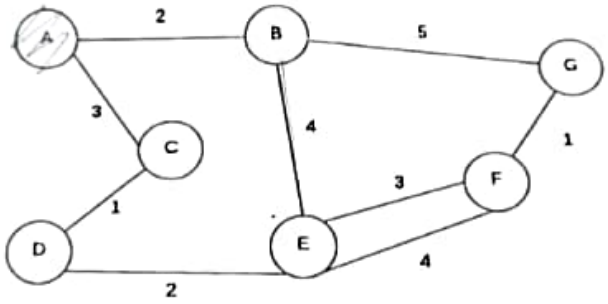
(a) Write an algorithm to insert an element at the first location of a single dimensional array? Analyse the time complexity of this algorithm

(b) Evaluate the following postfix expression using stack
5 6 2 + * 1 2 4 / - 5 6 2 + * 12 4 / -

(c) Find the preorder, postorder and inorder traversals of the following binary tree



(d) Find the minimum spanning tree of the following graph using Prim's algorithm



UNIT-I

Q2 (a) Describe the array data structure? (2+4+4=10)
(b) Write an algorithm/program to delete the last occurrence of an item from an array
(c) Analyse the complexity of the above algorithm

Q3 (a) Let *a* be a two dimensional array of size 8X12. If the first element of the array is stored at location 5050, find the address of element a[5][6], considering that matrix is stored in (3+5+2=10)
(i) Row Major Ordering
(ii) Column Major Ordering
Assume that each element in the array require one memory location
(b) Write an algorithm/program to find the sum of the elements on the minor diagonals of a square matrix
(c) Analyse the complexity of the above algorithm

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UNIT-II

Q4 (a) Write an algorithm/program to delete kth element from a circular linked list (5+5=10)

(b) Write an algorithm to insert an element before the given element in a singly linked list

Q5 (a) Define stack and queue data structures? (2+4+4=10)

(b) Implement a stack using linked list

(c) Implement queue using two stacks.

UNIT-III

Q6 (a) What is a B Tree? (2+8=10)

(b) Perform the following operations in sequence on an initially empty B-Tree of order 4

Insert 32, insert 54, insert 72, insert 89, insert 45,
insert 12, insert 38, insert 56, insert 10, insert 77,
insert 66,Delete 45, Delete 54, Delete 72, Delete 12, Insert 2

Q7 **(a) What is an AVL tree? Write algorithm/program for performing various AVL tree rotations** (6+4=10)

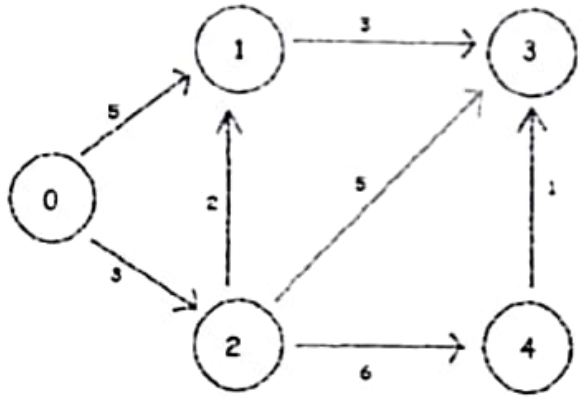
(b) Perform the following operations in sequence on the AVL tree in question 1 (c)

Insert 2, Insert 11, Delete 8, Delete 12

UNIT-IV

Q8 (a) Explain Dijikstra's single source shortest path algorithm? (5+5=10)

(b) Using Dijkstra's algorithm find the shortest path from vertex 0 to all other vertices in the following graph



Q9 (a) What is hashing? Explain the various collision resolution techniques? (6+4=10)

(b) Create a hash table of size =10 with the following keys
17,9,34,56,11,4,71,86,39,49

(i) Using Chaining

(ii) Using Linear Probing